

Incident Response Report

SOC Monitoring & Incident Response Simulation

1. Executive Summary

This assessment simulated a Security Operations Center (SOC) workflow using Splunk to ingest and analyze log data. The goal was to detect suspicious events, classify incidents by severity, and provide remediation and communication steps.

The exercise demonstrates SIEM alert monitoring, incident triage, and reporting, aligned with industry SOC practices.

2. Environment Setup

- ▶ Platform: Splunk Free Edition (local instance on Kali Linux)
- ▶ Dataset: simulated SOC logs (includes login events, IP connections, malware detections)
- ▶ Tools Used:
 - Splunk (log ingestion, dashboards, queries)
 - Excel (Alert_Classification.xlsx) for severity categorization
 - Screenshots of Splunk dashboards

3. Findings

3.1 Suspicious Alerts Identified

From Splunk searches and dashboards:

- 1. Multiple Failed Logins**
 - Several failed login attempts for users (admin, wiener) from unusual IPs.
 - Indicates brute-force attempts.
- 2. Unusual IP Access**
 - Access attempts from IP ranges outside expected geography.
 - Could indicate reconnaissance or compromised credentials.
- 3. Malware Detection Events**
 - Alerts showing malware signatures triggered on endpoints.
 - Requires immediate containment.
- 4. Privilege Escalation Attempt**
 - Logs show suspicious sudo or root activity after failed logins.

4. Incident Classification

(from Alert_Classification.xlsx)

High Severity

- ▶ Malware detection alerts
- ▶ Multiple failed logins (possible brute force)

Medium Severity

- ▶ Unusual IP logins

Low Severity

- ▶ Single failed login attempts

5. Impact Assessment

- ▶ Brute Force Attempts → risk of account compromise.
- ▶ Malware Alerts → endpoint may be infected, leading to lateral movement.
- ▶ Suspicious IP Access → may indicate attacker probing systems.

6. Recommended Actions

Containment

- Block malicious IP addresses at firewall.
- Isolate infected hosts.

Eradication

- Run antivirus/EDR scans.
- Reset compromised user credentials.

Recovery

- Monitor restored systems closely.
- Harden SSH login (limit attempts, enable MFA).

Prevention

- Enable Geo-IP filtering.
- Implement SIEM alert playbooks for failed logins.

7. Dashboard Evidence

- Login attempts per user (bar chart).
- Failed logins per IP (bar chart).
- Malware alerts over time (timechart).
- Incident summary (pie chart).